

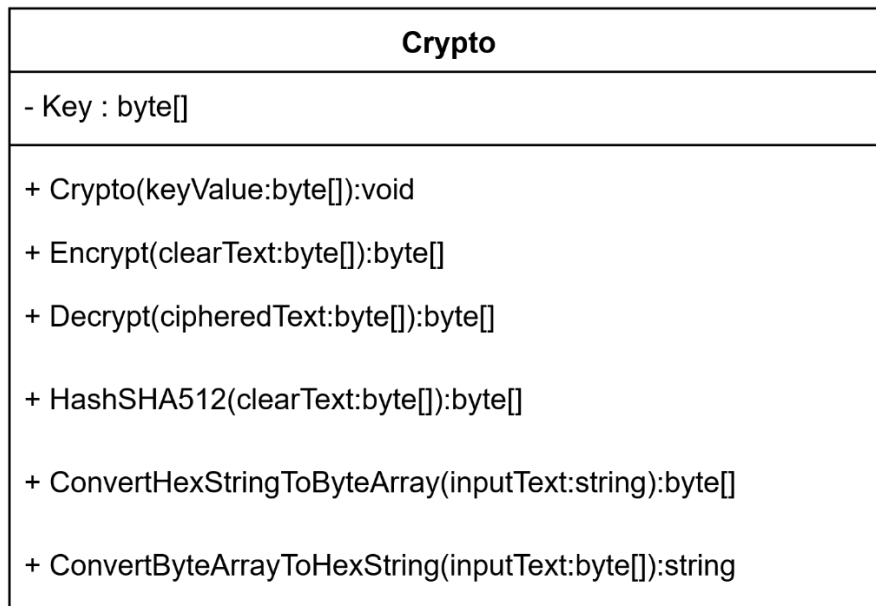
2-) Create a .Net Core WinForm application for basic encryption and hashing operations.

Application design should have only one form with following components.

3 Text box for entering the Key and Input value and displaying the Output value.

3 buttons to select the operation. (Encrypt, Decrypt, Hash)

Create a Cryptography class as in the UML diagram below.



For 3DES operations “System.Security.Cryptography.TripleDES” , for SHA512 operation “System.Security.Cryptography.SHA512” class can be used.

Suggested class documentations in below links.

<https://learn.microsoft.com/en-us/dotnet/api/system.security.cryptography.tripledes?view=net-8.0>

<https://learn.microsoft.com/en-us/dotnet/api/system.security.cryptography.sha512?view=net-8.0>

For 3DES operations use the followings,

-KEY: Value entered by user

-IV: 0000000000000000

-CipherMode: ECB

-PaddingMode: Zeros

Sample 3DES key: E037624AB97576042FCB68CB10D6B5EC